25900414 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCGEO200 - MINERALOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer in 60 words

Answer any **7** questions

Each question carries **2** marks

1. Draw the crystallographic axes of triclinic system. [K] / [CO1]
2. Define paramagnetic minerals. [K] / [CO2]
3. Name any four parts of a petrological microscope. [K] / [CO3]
4. Name the type of minerals that show double refraction. [U] / [CO3]
5. Name one polymorphic mineral pair of carbon. [K] / [CO2]
6. Describe suiphide group of minerals with example [U] / [CO4]
7. Why are amphiboles less stable at high temperatures compared to pyroxenes? [U] / [CO5]
8. List two important ore minerals of iron from the oxide group. [U] / [CO6]

[2x7 = 14]

Part B

Short notes in 250 words

Answer any **3** questions

Each question carries **8** marks

9. Describe how streak offer a more dependable clue to mineral identity than surface colour. [U] / [CO2]

10. Compare rotation and roto-inversion symmetry. [U] / [CO1]

11. Describe the phenomenon of total internal reflection and its significance in optical mineralogy. [U] / [CO3]

12. Discuss the role of halides in the chemical industry. [A] / [CO6]

13. Compare the geometries of triangular and cuboctahedral coordination polyhedra. [U] / [CO4]

[8x3 = 24]

Part C

Essays in 400 words

Answer any **1** questions

Each question carries **12** marks

14. Compare the retardation values and interference colour effects of quartz wedge, gypsum plate, and mica plate and classify their applications in mineral identification. [U] / [CO3]

15. Explain the polymorphism of Al_2SiO_5 minerals and their significance in metamorphic petrology with diagrams. [An] / [CO5]

[12x1 = 12]